

FoodSaver: A Prototype for Reducing Household Food Waste

Abstract— Food waste remains a major sustainability challenge, and in Saudi Arabia the issue is pronounced at the household level. Practical causes include weak inventory tracking, expiration dates being forgotten, and purchasing decisions that are not consistently aligned with what is already available. Although awareness has grown, tools that translate this awareness into everyday behavior through low-effort, timely support are still limited.

This paper introduces FoodSaver, a mobile application prototype designed to reduce household food waste through four integrated features: a visual expiration-date calendar, a category-based inventory system, an ingredient-driven recipe recommendation engine, and a location-aware donation module. The prototype was developed using a User-Centered Design (UCD) framework, with Fogg's Behavior Model (FBM) used as a persuasive design layer to support motivation, ability, and timely triggers. Development and evaluation followed an iterative process that included semi-structured interviews ($n = 10$), an online survey ($n = 35$), heuristic evaluation against Nielsen's usability principles, and usability testing using three Think-Aloud sessions.

Overall, the results indicate that FoodSaver can bridge the gap between knowledge of food waste and actionable household behavior by making food management easier and more immediate.

Keywords—food waste; mobile application prototype; user-centered design; Fogg Behavior Model; Saudi Arabia; sustainability

I. INTRODUCTION

Food waste is one of the most persistent and least addressed sustainability challenges of the modern era. In Saudi Arabia, the problem is particularly acute, with households discarding food at rates substantially higher than the global average [6], [7]. This pattern is often the result of a combination of cultural practices, such as generous food

preparation for social gatherings, and behavioral habits like redundant purchasing caused by a lack of real-time inventory awareness [8], [9]. Because tracking perishable items relies almost entirely on human memory, items frequently expire unnoticed, leading to significant environmental and economic costs [8].

While recent years have seen a rise in public awareness regarding food waste, awareness alone has proven insufficient to drive meaningful behavioral change [10]. For individuals to adopt more sustainable routines, they require practical, high-utility tools that integrate naturally into their daily lives—tools that move beyond general information to provide active, timely support at the point of decision-making. Mobile applications offer a promising platform for this intervention, as they can consolidate features like expiration reminders, inventory logging, and recipe suggestions into a single, accessible interface [11]. However, many existing solutions address these dimensions in isolation, leaving a gap for integrated, context-sensitive tools specifically tailored to the Saudi Arabian household context.

To address this gap, this paper introduces FoodSaver, a smart mobile application prototype developed to streamline household food management. FoodSaver integrates four core capabilities: (1) a visual expiration calendar for proactive tracking; (2) a checkbox-based inventory system organized by food category; (3) an ingredient-based recipe suggestion module to utilize existing stock; and (4) a geolocation-enabled donation feature that connects users with local charitable organizations. The prototype was designed specifically for primary food decision-makers—including parents, homemakers, and university students—ensuring the features align with their specific logistical needs.

FoodSaver's development followed a UCD framework [1], which ensured that design decisions were grounded in real-world user needs through an iterative four-phase process. FBM served as a persuasive design layer, ensuring the prototype effectively combines motivation, ability, and timely triggers to encourage waste reduction [2]. The system's

efficacy was evaluated using Nielsen's usability principles [3] and Think-Aloud testing with representative participants. The remainder of this paper is organized as follows: Section II reviews relevant literature; Section III details the methodology and design framework; Section IV presents the results; Section V discusses findings; Section VI explores future work; and Section VII concludes the study

II. LITERATURE REVIEW

The reviewed papers examined the problem of food waste in Saudi Arabia from multiple perspectives, including behavioral, regulatory, environmental, and technical dimensions. Despite the differences in their approaches, all papers agreed on a clear gap between theoretical awareness and actual practice, which formed the comparative foundation for designing the FoodSaver prototype as a technical solution that directly responds to the problems identified in the literature.

A. Food Waste Problem

Research in [6] highlights that the lack of tracking and measurement tools represents a major barrier to food waste management, emphasizing that users' unawareness of their food inventory status and expiration dates leads to a loss of control over consumption. In response to this issue, FoodSaver addressed this point by adding a digital calendar for food inventory management, which allows users to record products and track their expiration dates in an organized manner. This calendar transforms the tracking process from an activity that relies on human memory into a visual system that can be referenced at any time, which aligns with the recommendations provided in [6] regarding measurement and monitoring.

On another note, [7] explains that one of the main causes of food waste is poor planning and impulsive buying, where edible food is discarded simply to make room for new purchases. FoodSaver addresses this problem through inventory management based on user-entered data, where users become aware of what they actually own before making a purchasing decision. This approach reduces repeated buying of the same items and limits unnecessary stockpiling. The calendar and product records also support advance planning, which was lacking in the traditional solutions discussed in [7].

Forgetting expiration dates is identified as one of the most common causes of food waste in households [8]. This study notes that relying on human memory to track a large number of products is inherently inefficient. In response to this problem, FoodSaver relied on a smart notification and alert system, where the prototype sends proactive alerts a suitable period before the expiration date. These notifications serve as a direct intervention mechanism that reduces the likelihood of food spoilage due to forgetfulness and also allows the user to make an early decision to consume, donate, or reuse the food, thereby translating theoretical recommendations into a functional prototype application.

The literature also highlighted a lack of knowledge about how to make use of surplus food and the social perception of leftover food as undesirable [8]. FoodSaver addressed this gap by creating an interactive community for suggesting recipes using surplus or near-expiry foods. This feature

allows users to exchange practical ideas for reusing food instead of discarding it, which shifts the negative view toward food surplus and promotes sustainable behavior through social learning and participation.

Regarding food donation, [6] identifies this practice as an effective solution to reducing waste but implicitly notes that the difficulty of reaching appropriate organizations may limit its application. FoodSaver tackled this challenge by adding a digital donation feature linked to the user's geographic location, allowing the user to identify their location and easily access nearby charitable organizations. In doing so, the prototype not only encourages donation but also facilitates its actual implementation and reduces the logistical barriers mentioned in [6], [9].

From a human-computer interaction (HCI) perspective, [10] emphasized that the gap between knowledge and behavior can only be addressed through interactive systems that reduce cognitive load (CL) and provide timely support. This is reflected in FoodSaver's design, which relies on notifications, feedback, and visual motivation such as rewards after donation, encouraging users to maintain positive behavior without imposing it. Additionally, integrating community interaction within the application helps create collective awareness, rather than being limited to isolated individual decisions, as supported by the finding in [10].

B. Food Waste in Saudi Arabia

Saudi Arabia has higher levels of food waste than the rest of the world [6], [7]. Consumer behavior and cultural customs, such as over-purchasing, improper food handling, and misinterpretation of expiration dates, are the primary causes of this issue [6], [9]. Due to societal and cultural traditions, food is frequently prepared in large quantities in many households, which raises the possibility of waste. Food waste also leads to major environmental issues such as resource depletion, increased greenhouse gas emissions, and wasteful use of energy and water [8], [9].

Even while people are becoming more conscious of the detrimental effects of food waste, there is still a noticeable disconnect between knowledge and actual action, suggesting that awareness is insufficient to successfully reduce waste [10]. Many people are aware of the problem, but they do not have the means to manage their food effectively. As a result, it has been suggested that technology-based solutions, especially mobile prototypes, are useful tools for promoting behavior change and enhancing food management [11], [12]. These prototypes can help users keep track of expiration dates, organize their food, and make better consumption decisions.

By offering features like food tracking, reminders, and recipe recommendations to promote more sustainable practices, the FoodSaver prototype embodies these strategies. The prototype seeks to assist users in reducing waste, enhancing their everyday routines, and promoting environmental sustainability by integrating these elements into an intuitive interface.

C. Technology Solutions

Recent technology-based studies have demonstrated the effectiveness of digital solutions in reducing household food waste through intelligent food management, inventory

tracking, and recommendation systems. Exspiro, a mobile application for food sustainability, highlighted the importance of food item tracking, timely updates, and statistical feedback to support sustainable food consumption behavior and improve household food management practices [16]. Similarly, the artificial intelligence (AI)-assisted food management model proposed by Alfiora and Gumulya applied behavioral analysis and AI techniques to support food consumption management among urban households. The study also emphasized the importance of interviews, questionnaires, and usability testing in developing user-centered food waste reduction systems [17].

Recipe recommendation technologies represent another important technological direction for minimizing household food waste. Koreddi et al. proposed a multi-modal deep learning framework that utilizes ingredient-based knowledge extraction, natural language processing (NLP), and computer vision techniques to improve recipe recommendation accuracy and ingredient recognition [18]. In addition, Chotipant introduced a semantic-based recipe recommendation approach that addresses the limitation of exact ingredient matching by suggesting substitute ingredients based on semantic characteristics such as taste, texture, and smell [19]. These intelligent recommendations and approaches support users in making better cooking decisions and encourage the use of available ingredients instead of discarding them.

Recent inventory and grocery management technologies also contributed to improving food organization and reducing waste. Jagatheesan et al. proposed an intelligent grocery management system using Internet of Things (IoT) technologies for personalized shopping and inventory management [20]. The system focused on monitoring household inventory and supporting users in managing food products more efficiently through smart tracking methods.

Although these studies provide valuable technological solutions, many of them focus on isolated functionalities such as inventory tracking, artificial intelligence-based food management, recipe recommendation, or grocery monitoring separately. In contrast, the proposed FoodSaver prototype integrates multiple food management functions within one UCD. The prototype combines inventory tracking, expiration-date reminders, calendar-based food monitoring, ingredient-based recipe suggestions, donation support, and community interaction within a unified interface. This integrated approach enables the prototype to address multiple causes of household food waste rather than focusing on a single technological feature.

Furthermore, the proposed prototype emphasizes usability and accessibility through simplified interaction methods and reduced cognitive load. Features such as predefined ingredient categories, checkbox-based ingredient selection, visual expiration tracking, and organized recipe suggestions support more efficient user interaction and align with everyday household food management practices. Therefore, the FoodSaver prototype builds upon previous technology-based solutions by combining practical food management functions with user-centered interaction design principles.

Paper	Inventory Tracking	Recipe Recommendation	Donation Support	Community Interaction	Integrated Solution
[11]	✓	X	X	X	✓
[12]	✓	X	X	X	✓
[13]	✓	X	X	X	X
[16]	✓	X	X	X	X
[17]	✓	X	X	X	X
[18]	X	✓	X	X	X
[19]	X	✓	X	X	X
[20]	✓	X	X	X	X
Food Saver	✓	✓	✓	✓	✓

TABLE I. Technological features comparison.

Table I summarizes the key differences between the reviewed technology-based solutions and the integrated features of the proposed FoodSaver prototype.

III. METHODOLOGY & MATERIALS

A. Project Overview

FoodSaver is a smart home-based mobile application prototype designed to reduce household food waste through four integrated features: (1) visual expiration tracking via a calendar interface; (2) checkbox-based food inventory management organized by predefined food categories; (3) ingredient-based recipe suggestions with match percentage scoring; and (4) a community recipe-sharing platform with expert home-cook validation. The project targets Saudi households and primary food decision-makers within the home.

The project was iteratively developed across four phases: within the CIS 422 course. The final deliverable is a high-fidelity prototype built in Figma (Version 116, desktop application, accessed 2025–2026), encompassing 15+ interconnected screens covering the full user journey from onboarding through donation.

Key design improvements applied across phases: relocating the expiry alert to the top of the Home screen; adding a pre-submission review step before donation confirmation; replacing grey expiry warning colors with high-contrast indicators; greying out past calendar dates; showing missing recipe ingredients with a shortcut to inventory; adding a ‘Mark all as read’ button to Notifications; correcting ‘India’ to ‘Hindi’ in the language list; adding confirmation dialogs before account deletion and logout; and introducing a complete login and sign-up flow for new users.

B. Design Framework

FoodSaver was developed using the UCD framework [1], an iterative methodology that places user needs, behaviors, and limitations at the center of every design decision throughout the entire development lifecycle. UCD was selected for three reasons. First, food waste is fundamentally a behavioral problem embedded in daily household routines; a system built on assumptions rather than real user insight would be unlikely to change those routines. Second, UCD’s iterative structure—observe, design, evaluate, refine—maps naturally onto the four-phase structure of this project. Third,

the Saudi household context requires cultural sensitivity that only direct user engagement can supply.

FBM [2] served as the persuasive design layer complementing UCD. FBM states that a target behavior occurs when motivation, ability, and a trigger are simultaneously present. Each factor is explicitly addressed in FoodSaver's design:

- **Motivation** — reward badges ('Hero', 'Supporter') awarded immediately after donation completion, making prosocial behavior intrinsically rewarding.
- **Ability** — checkbox-based inventory entry requiring no typing, minimizing cognitive and physical effort for primary target users.
- **Trigger** — automated push notifications and a visual expiration calendar that surface time-sensitive information before food expires.

UCD and FBM combined work to make FoodSaver not only usable but also persuasive – easy enough for consistent use and designed to motivate proper behavior when necessary.

C. Data Collection

Two complementary methods were employed to achieve both depth of understanding and breadth of coverage across the target user population.

1. Face-to-Face Semi-structured Interviews:

Ten interviews lasting between 10 and 20 minutes each were conducted with respondents who had provided an informed consent form and thus demonstrated willingness to participate (interviews took place on 24–27 February 2026). Interviewees were adults aged 18 years or above and who engage in grocery purchasing/cooking. The participants were from a diverse background to represent different perspectives of the household. They were university students, parents, working adults. Questions included food storage practices, grocery shopping habits, main reasons for waste, awareness of expiration date and receptivity to reminder-based mobile technology. Written notes were taken throughout; audio recordings where consent was given.

2. Online Survey (Google Forms):

The qualitative interview analysis identified patterns that were then quantified using a structured online survey which was distributed. The survey was completed by 35 people in the primary target demographic and included: frequency of expiration tracking, preferred data input methods, desirable application features, and demographic information. The survey was conducted after the interviews so the questions could be based on actual language used by users instead of the researchers' assumptions. The results directly informed FoodSaver's core design decisions, as illustrated in Figure 1.

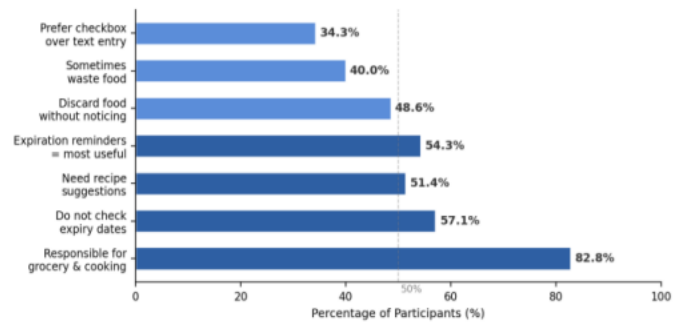


Fig. 1. FoodSaver user study findings (n = 35). Dark bars ($\geq 50\%$) drove core design decisions.

D. Interview Participants

Ten participants provided written informed consent and completed face-to-face interviews between 24–27 February 2026. All participants were adults actively involved in household food management. Table II lists the participants.

#	Name	Date
1		26/02/2026
2		24/02/2026
3		24/02/2026
4		25/02/2026
5		25/02/2026
6		25/02/2026
7		26/02/2026
8		26/02/2026
9		26/02/2026
10		27/02/2026

TABLE II. Interview Participants (Phase 2).

E. Usability Test Participants

participants representing primary target segments were recruited for Think-Aloud usability testing in Phase 4 to validate the prototype's functional flow across diverse user profiles. Table III provides a representative sample of these profiles, highlighting key demographics such as university students representing young adults managing personal inventory and household managers who serve as primary decision-makers.

Name	Profile	Age	Relevance
User 01	University Student	18–24	Represents young adults using food apps for convenience & sustainability
User 02	University Student	18–24	Represents users managing personal food inventory and meal planning
User 03	Housewife & Mother	35–44	Primary target user — household food manager

TABLE III. Representative User Profiles for Usability Testing (Phase 4).

F. Development Phases

Phase 1 — Pre-Design Analysis: Four competing systems were reviewed including NoWaste and FridgePal (inventory management), Yummly (recipe recommendation) and Too Good To Go (surplus food redistribution). No one has all four capabilities in one interface. Through a task

analysis we identified six key user tasks: (1) logging inventory; (2) scheduling expiration; (3) smart notifications; (4) recipe discovery; (5) community sharing; and (6) expert recipe validation.

Phase 2 — Initial Paper Prototype: Based on survey data, five core screens were designed and justified: Home screen: expiry warnings surfaced (48.6%) Inventory: checkboxes used (34.3%) Calendar: passive date visualizations (57.1%) Recipes: match percentages (51.4%) Community: verified/pending badges All screens connected via a persistent bottom navigation bar.

Phase 3 — Design Refinement Usability friction points resolved via Shneiderman's Golden Rules [5] Key changes: expiry alert moved to top; instructions repositioned; save/like features added to recipes; badge legends added inline. High-fidelity prototype built in Figma (v116) with 15+ screens.

Phase 4 — Evaluation: A peer group applied Nielsen's 10-heuristic evaluation [3] on the Figma prototype. Three Think-Aloud usability sessions followed (Table III). Sessions lasted 10–15 minutes using a paper prototype in a smartphone-style frame. Data was captured through written observational notes.

IV. RESULTS

A. Evaluation Overview

The purpose of this section is to describe the results obtained during the final evaluation process of the FoodSaver prototype. In order to achieve a detailed analysis of user experience, this evaluation included three approaches:

- **Quantitative Analysis:** Assessing the system's efficiency based on completion and performance rates.
- **Qualitative Analysis:** Analyzing subjective user experience and satisfaction by means of tests with target users (students and housewives).
- **Heuristic Analysis:** Conducting a thorough examination of the system's interface by peer evaluators from another project team, relying on Nielsen's Ten Usability Heuristics as a benchmark to solve any interface issues from an outsider's point of view.

B. Quantitative Results

Here are the results from representative participants (User 01, User 02 and User 03) while carrying out tasks across the primary application interfaces, including the **Home Screen (Fig. 3)**, **Food Inventory (Fig. 4)**, **Expiry Calendar (Fig. 5)**, **Recipe System (Fig. 6)**, and **Donation Dashboard (Fig. 8)**:

1. Task Success rate:

Participant	Task	Time Spent	Observation
User 01	Complete donation process.	<20s	Strength: Reward badges motivate donation behavior. Issue: No pre- submission review step.
User 02	Calendar tracking and validity alerts.	45s	Strength: Red highlight on expiry dates is clear. Issue: Expiry warning color too subtle; past dates ambiguous.

User 03	Search for a recipe based on stock.	30s	Strength: Recipe suggestions are intuitive and accurately based on available stock. Issue: Missing ingredients are not shown.
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TABLE IV. Task Success Rates.

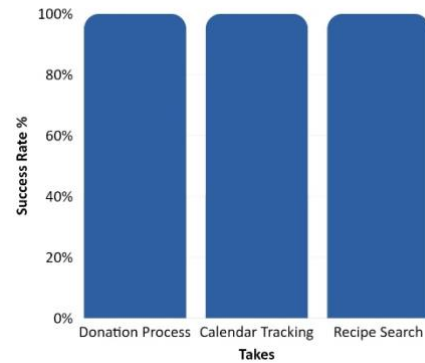


Fig. 2. Success rate.

As shown in **Fig. 2**, all three usability tasks achieved a 100% success rate, indicating that participants were able to complete each task without errors.

2. Performance Analysis:

1. Efficiency & Time Analysis:

Participants labored for a greater or fewer period based on: contrast in time they had, explicitly the following nature of duties that performed while tested with the FoodSaver system.

- **High efficiency (User 01):** The donation system maintained the fastest time of less than 20 seconds while confirming that the whole process does not distract and is clear on how the users path should be (see **Fig. 8**).
- **Long time (User 02):** Dealing with the calendar and alerts took 45 seconds; this explains why it needed more time to distinguish between alerts and dates due to visual interference (see **Fig. 5**).
- **Average time (User 03):** Looking for recipes and choosing ingredients: 30 seconds; great time for a task focused on content selection Alternates (see **Fig. 4** and **Fig. 6**).

2. Strengths & Issues:

We categorized observations based on how well participants performed at their assigned tasks:

1. Donation Task (User 01):

- **Strength:** Reward Badges proved highly effective in motivating donation behavior, transforming the process into a fun and rewarding experience (**Fig. 8**).

- Issue: The absence of a "Pre-submission review" step could lead to unintentionally sending incorrect donations.

2. Calendar and Alerts Task (User 02):

- Strength: The red alerts for expiry dates were extremely distinct and simple to determine (Fig. 5).
- Issue: A few warnings used subdued colors and therefore we didn't notice them initially, plus the format of past dates on the calendar was unclear.

3. Recipe Search Task (User 03):

- Strength: Effectiveness across ingredient tags top of the screen → these ingredients were easier to keep in mind and had lesser cognitive costs associated with them (Fig. 6).
- Issue: The "Missing ingredients" option was not visible in the recipe card making it impossible to plan your shopping.

C. Qualitative Results

This section focuses on the subjective experiences and feedback from the previously mentioned representative participants, highlighting what they said and felt during their interaction with the system:

User Satisfaction & Feedback:

We received qualitative feedback from the sessions, and they were largely positive too; people felt like features of the prototype matched what they used daily.

- Motivation by Gamification (User 01): The badge system (e.g., "Hero" and "Supporter") was her favorite feature. That was a big shift from food donation being a tedious task to one where you get "points for fun," and that would probably make her use the prototype more often, she said (Fig. 8).
- Practicality for the Housewife (User 02): Noted that the calendar reminders are a "godsend" for running a busy household. However, she said, it should be more visually obvious which dates the user has missed and which ones they have not, to avoid inadvertently clicking on one from last week (Fig. 5).
- Cognitive Load (User 03): pointed out that the recipe proposal feature is very "smart". She liked the ingredient tags, saying that she did not need to monitor what she picked, which lowered her cognitive demand when preparing a meal (Fig. 6).

D. Heuristic Evaluation Results

A Heuristic Evaluation was conducted on the FoodSaver prototype by peer evaluators from a different project group. This evaluation applied Nielsen's ten usability guidelines to identify interface weaknesses from an external perspective. The table below highlights the

identified issues and the subsequent design improvements implemented:

Principle	Identified Weakness	Design Improvement (Iteration)
1. Visibility of System Status	Old notifications persisted in the list, causing clutter and confusion.	Integrated a 'Mark all as read' button to allow users to clear and update their status.
2. Match between System & Real World	Language menu incorrectly listed countries (e.g., India) instead of language names.	Updated the menu to display correct language names (e.g., Hindi/English).
3. User Control & Freedom	Critical actions like 'Delete Account' were executed immediately without confirmation.	Implemented a Confirmation Pop-up (Yes/No) to prevent accidental data loss or logout .
4.Consistency & Standards	Spelling errors in interface labels (e.g., 'commuinity') undermined professionalism.	Conducted a Linguistic Audit and corrected all typos across the prototype pages.
5. Error Prevention	Past calendar dates were visually identical to future dates, leading to invalid selections.	Applied a Faded/Grey style to past dates to visually communicate they are non-selectable (Fig. 5).
6.Recognition Rather Than Recall	No visual link between suggested recipes and the user's actual inventory.	Added Color-coded indicators for registered/unregistered ingredients within the recipe view.
7.Flexibility & Efficiency	Navigating long language lists was tedious for users.	Integrated a Search Box within the dropdown menu to allow quick filtering of options.
8. Aesthetic & Minimalist Design	UI layering issues caused menus to overlap with header lines, creating a cluttered look.	Adjusted Layer Depth (Z-index) and spacing to ensure clean interface transitions.
9. Help Users with Errors	The system lacked feedback when users interacted with invalid (past) calendar dates.	Developed Error Messages that appear for 5 seconds, explaining the failure and recovery steps .
10. Help & Documentation	New users had no clear starting point or account creation process.	Designed a standardized Onboarding flow including Login and Sign-up screens for new users.

TABLE VI. Summary of heuristic evaluation findings and design iterations.

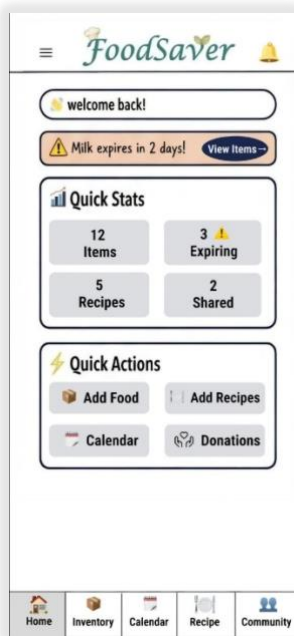


Fig. 3. Home Screen - An interface that displays quick statistics for inventory and expired items with quick access buttons.

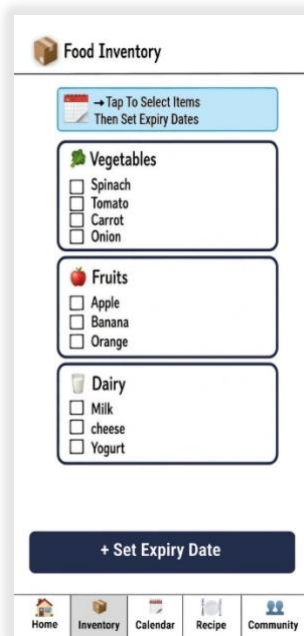


Fig. 4. Food Inventory - A list to organize foods by category (vegetables, fruits, dairy) and track their expiration dates.



Fig. 5. Expiry Calendar - An interactive calendar that identifies product expiry dates to regulate consumption.

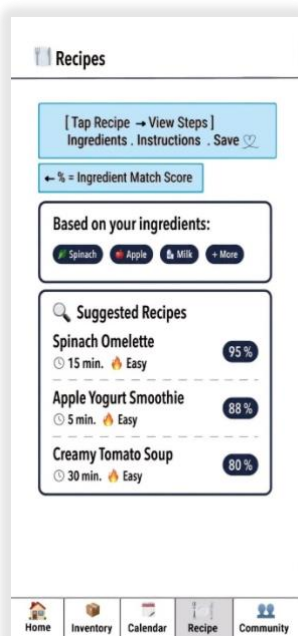


Fig. 6. Recipe System - Suggestions for smart meals that rely primarily on available ingredients to reduce waste.

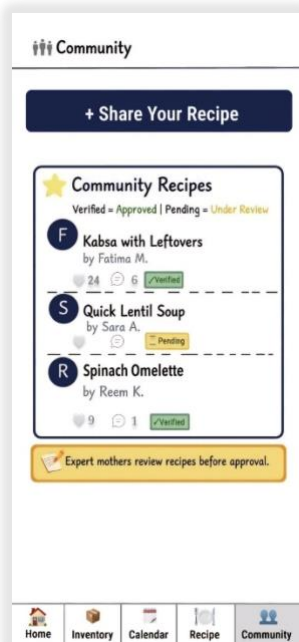


Fig. 7. Community - A platform for sharing user recipes and promoting a culture of food sustainability.

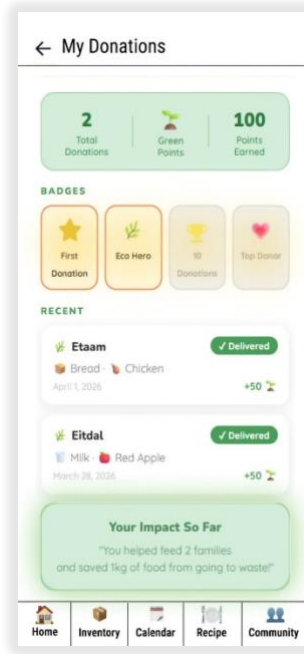


Fig. 8. My Donations - A personalized dashboard showcasing donated items, gamification rewards, and real-time impact tracking to promote food sustainability.

IV. DISCUSSION

The final evaluation confirms that FoodSaver effectively achieves its core objectives with a 100% task success rate across all tested scenarios. The results indicate a strong balance between functional utility and user engagement:

Gamification Success: The positive feedback regarding "Reward Badges" confirms that the motivation-based design, inspired by the Fogg Behavior Model, successfully encourages sustainable social behavior.

Refining Efficiency: While task completion times were efficient (averaging under 45 seconds), the testing highlighted a need for better information scent—specifically, displaying "missing ingredients" within the recipe view to streamline the user's planning process.

Error Reduction & Heuristics: The transition from subtle grey alerts to high-contrast visual cues, along with the integration of confirmation steps (as detailed in Table VI), significantly reduces the risk of user slips and accidental data loss.

Overall, these refinements ensure that the FoodSaver prototype is not only highly usable but also provides a professional, intuitive, and rewarding experience for university students and housewives alike.

V. FUTURE WORK

Taking into account the analysis done, future work will continue to improve FoodSaver by:

- Utilizing AI methods for analysing the consumption patterns of households and offering personalized recommendations for minimizing unnecessary purchases.
- Including donations from an automated system linked directly to recognized charitable organizations according to one's geographic location.
- Providing a barcode scanner for easier data input and less physical labor involved in recording items.
- Creating a dashboard for users to monitor their savings in terms of food and other resources.

VI. CONCLUSION

The FoodSaver proposed prototype represents a total technological approach towards solving the issue of food wastage in the Kingdom of Saudi Arabia. The app distinguishes itself from other solutions as it incorporates the following four aspects: inventory control, expiration reminders, recipe recommendations, and community donations. The approach chosen for designing the product interface, namely the UCD method, was successful in developing interfaces that decreased cognitive load while taking into account the purchasing behavior of Saudi customers. The findings obtained clearly show that visual tracking combined with social motivation performs better than conventional education efforts.

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